PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	The Efficacy of Virtual Reality-Based Interventions for Breast Cancer Patients Symptom and Rehabilitation Management: A Systematic Review and Meta-Analysis
AUTHORS	Zhang, Huayi; Xu, Hui; Zhang, Zhen-xiang; Zhang, Qiushi

VERSION 1 – REVIEW

REVIEWER	Anothaisintawee, Thunyarat Ramathibodi Hospital, Mahidol University, Department of family medicine
REVIEW RETURNED	29-Jun-2021

GENERAL COMMENTS	The authors performed systematic review and meta-analysis of the efficacy of VR in breast cancer patients. The research question is interesting but there are many limitations that are required authors to correct and address, especially for review method, statistical analysis, and the way to present the results as follows Abstract 1. What is the definition of case-controlled trial? 2. Inclusion and exclusion criteria stated in the abstract are not clear. Inclusion criteria should be based on review question (PICO). 3. Authors said that qualitative summary was used but why they present the results from meta-analysis that is quantitative summary. 4. “four sections of cognitive function, and the range of motion of shoulder (details attached in this paper)” in the abstract (page 4, line 6-9) is not clear. What does it mean for this sentence? 5. It is not clear for the comparator. What is comparator of this study? Introduction 1. The term “new topic” in page 5, line 28 is not clear. What is the new topic? Which areas of breast cancer that the authors want to mention? 2. The evidence of VR in breast cancer patients should be included and discussed in the introduction to strengthen the rationale of conducting this study. 3. For the objective, the population in the study was not clear enough. I suppose that not all breast cancer patients can be prescribed for VR. Therefore, the authors should specified what types of breast cancer patients for instances breast cancer patients who were undergone MRM, are more likely to have the benefit from VR? Methods 4. What is the comparator of this study? The comparator should be included in the inclusion criteria.

	5. Measurement method of each outcome should be included in the section of types of outcome. 6. I suppose the authors used old version of ROB tool for quality assessment. It will be better, if they use the RoB-2 that is the most updated version. 7. Why the authors classified the quality of each domain as met, unmet, and unclear. This is not the standard for assessing and classifying the quality of study using RoB tool from crochrane handbook. 8. It is not clear for this sentence “Because the participants of this intervention are all patients, it is theoretically difficult to be double-blinding.”. I think that unblinding is resulted from the nature of VR that cannot be blinded. 9. I do not recommend using score for judging high and low quality of evidence. 10. What is the definition of descriptive analysis that the authors mentioned in measurement of treatment effect? Is it the standard term or not? 11. What is the definition of mean difference? Does it mean difference between intervention and comparator or mean difference between pre and post treatment of the same intervention? 12. What was the method used for calculating weight in the meta-analysis? 13. “we considered heterogeneity among studies, but according to clinical judgment, there was consistency among studies, and effect size needed to be pooled, so we selected a random effect model” This sentence is very confusing. What did you men for this? You pooled only the outcome that did not have the heterogeneity between studies right? If yes, why you used random effect model for pooling the results instead of fixed effect model. 14. If the authors found heterogeneity between studies, sours of heterogeneity should be explored. 15. There was not publication bias assessment. Results 1. The terms that you used in Table 2 are very confusing such as VG, VR, VRG. These are the same meaning or not? If they are the same, please use consistent term. 2. I think it is not appropriate to said that your scoring method for risk of bias assessment is better than the standard version. You should read the manual of RoB in cochrane handbook carefully why the developers of RoB do not recommend or use scoring system for grading the quality of study. 3. The characteristics of participants should be included in Table 1 such as mean age, % female, disease staging, baseline QoL, depression score. 4. For rang of motion outcome, why 95% CI of MD of extension was not significant but P-value was less than 0.05? 5. Pre-post studies should not be included in the meta-analysis because this study type has no comparator. Discussion 1. Authors should state about strength and limitation of this study 2. What is the clinical implication from the findings of this study? For example what types of patients that are recommended for using VR?

REVIEWER	Schoenenberg, Katrin Bergische Universitat Wuppertal, Clinical Psychology and Psychotherapy

REVIEW RETURNED	02-Jul-2021

GENERAL COMMENTS	The article describes a systematic literature research and meta-analysis on studies using virtual applications for improving physical and psychological health of breast cancer patients. More precisely, the outcomes anxiety, depressiveness, fatigue, extend of pain, different cognitive functions and different motor ranges are investigated. The topic of the article is highly relevant. The introduction gives the reader a good overview on all important information. Regarding the methodological approach, I'd recommend some revision. As a first point, the model type (fixed or random) used in a meta-analysis should be based on theoretical consideration (Cuijpers, 2016). Due to very diverse study designs, using a random effect model could be the better choice for all analyses. Furthermore, it remains unclear whether the pre versus post change or the difference of VR versus control treatment/group is analysed. Since you include different study designs you need to be very clear about this aspect. In the forest plots you write "favours VR" and "favours control" and "experimental" and "control" even though some studies did not have a control group (House et al., 2016, included in six analyses; Chen et al. 2019; included in four analyses). If you'd like to include studies without a control group you can analyse pre versus post effects only. In this case, you need to be careful with discussing differences of VR in contrast to control treatments. Based on the statistical results you can interpret the effect of the VR treatment only. One solution could be to split the analysis clearly into a review part, in which you describe the outcomes of the RCTs, and a meta-analytic part, in which you include all studies investigated. Accordingly, in the discussion you need to point out which result is based on the review and which one is shown in the meta-analysis. If you decide for this approach, please specify further which sub-dimension of the SV-POMS (Chirico et al., 2019) and which score of the STAI (Mohammad et al., 2018) were reported and used for the analysis. In the depression analysis combined mean differences are used only. However, the two studies investigated depressive symptoms based on different questionnaires (according to Table 2: House et al.: BDI-II; Chirico et al. SV-POMS). I highly recommend to use SMD or some other effect size estimations for all analyses. A second solution would be to remove the meta-analysis completely. Some authors question meta-analysis based on pre-post data (Cuijpers et al., 2017). Since the number of studies included is very small you may question how meaningful the results are. In case you decide to keep the meta-analytic part, please revise the discussion and conclusion. Please clearly describe its limited validity and please point out that the results for some outcomes did not reach statistical significance (assuming that pre vs. post values of the VR group were analysed here). Furthermore, I recommend to change the word "depression" to "depressiveness" or "depressive symptoms" in the entire paper. The measures (BDI-II and SV-POMS) do not assess a diagnosis but only the severity of depression.

	Please add a general discussion including the following limitations of research so far: small number of studies, small sample sizes, very diverse programs, dosages and technologies. I'm not a native speaker myself. Nevertheless, my impression is that the paper might benefit from some further language check. Attached you find some further minor comments. In sum, I support a publication with the described changes. Further Minor Comments Figure 1, Please indicate the excluded cases with (n =…) e.g. Subject (n = 849), Design (n = 302) Table 1. As I understand the table, putting a "+" sign implied that a criterion was met and accordingly there was "no incomplete outcome date" and "no selective reporting". It may be helpful to rename the columns. Furthermore, there might be a small mistake regarding the study by Jin et al (35). In the table you can find four "+" signs but the overall score is 5. 2.2.3 p.12 r.45 "by the Cochrane Handbook": Please insert the according reference (39) here. 2.2.3 p.13 r.9 ", and the others are not scored": Please describe in more detail which criteria were scored and which ones weren't. 2.2.3 p.13 r.11-15 "The total scores were 6 points, and more than 3 points could be included in this paper.": Do you mean the following: "…, papers had to achieve at least 3 point to be included in the review" References Cuijpers, P. (2016). Meta-analyses in mental health research. A practical guide. Amsterdam, the Netherlands: Pim Cuijpers Uitgeverij. Cuijpers, P., Weitz, E., Cristea, I. A., & Twisk, J. (2017). Pre-post effect sizes should be avoided in meta-analyses. Epidemiology and Psychiatric Sciences, 26(4), 364-368.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
Dr. Thunyarat Anothaisintawee:
Thank you for your comments on our paper. Without your guidance, we can't make such a promising direction and resolve the tricky question. We have nearly revised most questions mentioned in your email. We will make a point-to-point reply here to explain our ideas and thoughts.
Abstract
1. The definition of a case-controlled study is that our original purpose included this type of study in our paper. It's a study that compares two groups of people: those with the disease or condition under study (cases) and a very similar group of people who do not have the disease or condition (controls). Researchers study the medical and lifestyle histories of the people in each group to learn what factors may be associated with the disease or condition. However, we did not find relevant literature in our retrieval process, but we forgot to delete the content in the abstract, which is our negligence. We have deleted it.

2. For the inclusion criteria, we stated not clear, we referenced the paper format of BMJ Open, we took the PICO format in the abstract; it is more detailed.
3. We may not have well described the results of the qualitative analysis. We believe that Table 2 is the content of the qualitative description, so we have more detailed content of relevant literature.
4. Along with the Q3, we didn't clearly express our outcomes in summary, and we detailed the contents of the results in the revised abstract.
5. As for this question, we stated the comparator of this study in the revised abstract. The control group was given conventional breast cancer care.

Introduction

1. The new topic is that the number of breast cancer patients is rising. We have revised the statement of the sentence.
2. The evidence of VR in breast cancer patients is discussed on page5 line 13 to strengthen the study's rationale.
3. For the objective, the population in the study was not clear enough. Therefore, we consulted the relevant papers. Combined with the subjects included in the paper, we took the TNM grade as a reference and included the patients in stages 1-4 on the study page5.

Method

4. We have updated the comparator of the study in the types of intervention section on page 7. Types of interventions.
5. Measurement of each outcome is included in the section of Table 2. Due to the broad research scope of outcome indicators, we have described the relevant research tools in Table 2 in detail. It looks more refined.
6. Reviewer 1's suggestion is excellent. Yes, we used the old version of ROB before. We considered the suggestion and thoroughly discussed it, so we made the final decision to make a completed brand-new ROB-2 for study. You can find it in Figure2.
7. Like the Q6, we have updated it with ROB-2
8. Yes, we appreciate your idea. The original description of the unblind is very confusing. Unblinding has resulted from the nature of VR that cannot be blinded. We took the idea of reviewer 1.
9. We have taken the ROB-2 for quality assessment.
10. Since some studies cannot calculate the combined effect size, we have conducted all literature quality evaluations, so we only did a qualitative analysis to illustrate our results. However, the results of the qualitative analysis should be treated with caution.
11. The mean difference here mainly refers to the difference between the intervention group and the control group. When outcome measurements in all trials are made on the same scale, we choose to mean difference (MD) as a summary statistic in meta-analysis. It estimates the amount by which the treatment changes the outcome on average.
12. We used mean difference (MD) and standardized mean difference (SMD) to analyze continuous data. For the mean difference method, the standard deviations were used together with the sample sizes to compute the weight given to each study. For the standardized mean difference approach, the standard deviation is used to standardize the mean differences to a single scale and the computation of study weights.
13. Because the measurement tools may differ for the same outcome, we used a random-effects model.
14. We agree with the view of reviewer 1. We carefully analyzed the included studies, and there was heterogeneity with other studies. However, due to the limited number of articles, we could not conduct sensitivity analysis and subgroup analysis to find the source of heterogeneity. However, in the discussion section, we explained that heterogeneity might be caused by the disease stage, intervention time and frequency of patients, and various factors.
15. At the beginning of the study, we considered using a "funnel plot" to represent publication bias, but the number of included studies is small due to the language bias. Visual examination and statistical analysis of funnel plots have limited power to detect bias if the number of studies is small.

So, we give up making the funnel plot. We are very grateful to the reviewers, and we will add a detailed description of publication bias in limitations.

Results

1. VR stands for virtual reality, VG stands for virtual reality group, and VRG has the same meaning as VG, so we merged the VRG to VG.
2. As mentioned in the previous article, we have used ROB-2 to evaluate literature quality.
3. Thank you for your advice. We described the characteristics of participants in more detail in Table 2. We added the mean age, but limited by the included research; we had to choose the range of age of participants. All the participants are all female. We also added the disease stage by the TNM stage system. As for the baseline QoL and depression score, some studies don't offer them statistics, so we decided not to take them into the table.
4. Our negligence caused the original data error. Now we only choose abduction of upper limbs for analysis, and the results are promising.
5. We evaluated the before-after study and randomized controlled trials separately. We chose qualitative analysis for the types of hybrid trials to elaborate the results and clarified that the results should be treated with caution.

Discussion

1. In the first version of the submitted manuscript, we haven't fully discussed this study's strengths and limitations, and we realized this question. Thanks, reviewer 1 quoted this point, so we have fully discussed them in the revised manuscript on pages 24-26.
2. That's a good question. The ultimate goal of scientific research is to turn it into clinical practice. It also points out the direction for us to further explore its application. We discussed these findings. For example, it provides an early or medium-term breast cancer adjuvant for treatment or rehabilitation and is recommended for patients with mental symptoms. You can see it in the discussion.

Dear Dr. Katrin schoenenberg:

Thank you for your issues and proposed an executable modification. This does indicate the direction for our different designs and modification. We feel very grateful. In this email, we will specify the specific statement; according to your guidance, we have made the revision and our reasons.

We read the paper you mentioned (Cuijpers, 2016) very carefully, and the theory in this paper is significant for our studies. Due to diverse study designs, using a random effect model is a better choice for the analyses. So, we revised the effect model.

As for the different study designs in the forest plots, we didn't consider the favors [texts] before. The varied study designs are challenging for us, especially the pre-post study. Therefore, we deleted the before-after study in the meta-analysis. We only selected RCT for meta-analysis. We used the JBI tool for evaluation for other studies, but there was no combined effect quantity. We conducted a qualitative analysis.

For the sub-dimension of the SV-POMS and score of the STAI, we read the full text and got the right sub-dimension and score of the tool, and we showed them in Table 2. In the depressive outcome, we changed the calculation method. We chose SMD to analyze the difference in scale; statistical analysis was shown in the results part.

Our team thoroughly discussed your opinion. We would keep the meta-analysis part. As mentioned above, we only use RCT for meta-analysis.

We revised the discussion and conclusion, clearly described its limited validity, and pointed out that the results of some outcomes did not reach statistical significance. You mentioned the problem of depression. We didn't know the experts in this area, so we decided to adopt your opinion, modify the words, and carefully read the scales' handbook. Your suggestion is excellent; we have already modified all words.

In the discussion part, we considered your suggestion and analyzed and discussed the limitations of this paper, such as a small number of studies, small sample sizes, very diverse programs, dosages, and technologies.

You are correct; our mother tongue is not English. Maybe our English academic level has no shortcomings; we have worked hard to modify non-authentic and non-logical expressions. I hope you can satisfy all changes this time.

Further Minor Comments reply are as follows:
1. We took the PRISMA flow diagram format, indicated and updated detailed information in the chart. You can see it in Figure 1.
2. We used the evaluation method of ROB-2, which you can see in Figure 2.
3. We inserted the reference by the Cochrane handbook.
4. We used the ROB-2.
5. Same with Q2 and Q4.
In short, thank you very much for your valuable comments for this revision; this is my first English paper; thank you for your affirmation and support.

VERSION 2 – REVIEW

REVIEWER	Anothaisintawee, Thunyarat Ramathibodi Hospital, Mahidol University, Department of family medicine
REVIEW RETURNED	28-Dec-2021

GENERAL COMMENTS	Methods • The last search is November 2020. Therefore, the authors should update the search to identify additional relevant studies. • For interested participants, the authors included breast cancer patients with TNM I-IV stages, but why did they exclude the patients with distant metastasis in the exclusion criteria? • Applying I2>=50% was too risky for choosing fixed or random effect models. I suggest using I2>=25% instead of 50%.

REVIEWER	Schoenenberg, Katrin Bergische Universitat Wuppertal, Clinical Psychology and Psychotherapy
REVIEW RETURNED	16-Dec-2021

GENERAL COMMENTS	Below you find some last comments on the current version of the article. I have also attached some suggestions regarding the wording used (P = page, L = lines). The revision improved the article and, in particular, pointed out the limitations of the available literature. With the minor changes (see below and attached document) I'd support the publication of the article. Best regards, Katrin Schoenenberg // - P10.L5-8 "A random-effects model was applied to calculate the pooled results if I2 ≥ 50%; otherwise, a fixed-effects model was used." In your reply you write that you revised the methods and used a random effect model. If this is was the case, please adapt this sentence. - In Figure 4 "favors virtual reality" is plotted to the right, in Figure 3 and 5 it is on the left-hand side. For the reader, it would be easier

	to understand the graphs if you plot “favors virtual reality” to the same side in all Figures. - P16.L16-21 “At the end of the intervention, the depression symptoms scores of patients in the before-after study decreased by 8.3 compared to baseline data ($p < 0.05$), …” and P16.L37-40 “… pain scores in the before-after study decreased by 1.5 compared to baseline ($p < 0.05$).” Please indicate exactly what kind of scores you are referring to (8.3 and 1.5). Are they SMD scores? - P19.L35-38 “Due to technical reasons, we failed to conduct a combined effect size analysis, but VR is beneficial to the cognitive function of patients.” I don’t understand this argument. Some further suggestions regarding the wording (P = page, L = lines). - P3.L56 – 59 “There are few intervention studies completed so far, so we could not make an exact recommendation and conclusion.” -> “To date, there are few intervention studies, therefore, giving precise recommendation or conclusion is difficult.” - P5.L5 + L9 “adverse reactions” -> “adverse effects” - P5.L13-16 “may further cause the patient's cognitive dysfunction” -> “may further cause cognitive dysfunctions for the patient” - P10.L24-26 “There were no missing data of all these included articles” -> “There was no missing data for any of the included articles” - P12.L40-42 “These researches were conducted in the specialized ward and cancer center of the hospital.” -> “The research was conducted in specialized wards and cancer centers of hospitals.” - P12.L49 “These studies assessed…” -> “The studies assessed…” - P17+18.L59-4 “Our studies also included them in our studies; in each of the studies, the results showed that VR positively affected these symptoms.” -> “Our review also included the effect of VR on depressive symptoms. In both studies analyzed, the VR treatment had a reducing effect on depressive symptoms.” - P18.L16 “They performed MRI analysis showed distraction effectively suppressed pain processing in the brain, while placebo may have no significant effect on pain processing.” -> “Their MRI analysis showed distraction effectively suppressed pain processing in the brain, while placebo had no significant effect on pain processing.” - P18.L40 “significantly reduce the quality of life patients” -> “significantly reduce the quality of life of patients” - P19.L51-54 “To sum up, VR used the immersive experience to intervene and achieved positive results. These devices specifically included: wearable helmets, 3D glasses, somatosensory operating devices, smart displays, and computers. After these devices are connected, they can interact with the outside world in real-time. Subjects can interact cognitively or physically in a computer-generated virtual environment to achieve their expected results. An immersive tool changes patients'

	awareness in a virtual environment, provides a positive and pleasant experience, and improves intervention." -> "To sum up, in this context VR uses immersive experience to intervene and achieve positive effects for breast cancer patients. The devices which were used are: wearable helmets, 3D glasses, somatosensory operating devices, smart displays, and computers. Subjects can interact cognitively or physically in a computer-generated virtual environment in real-time to achieve their goals. An immersive tool changes patients' focus of attention, provides a positive and pleasant experience, and improves progress." - P19.L23-26 "VR has real-time solid subjects in the virtual environment." -> This sentence is unclear. - P22.L21 "This study has some limitations that need to be addressed, and we had better treat the results of this study with caution. First of all, the types of studies included were relatively mixed,…" -> "This study has some limitations that need to be addressed, and which is why we need to interpret the results with caution. First of all, the types of studies included were relatively diverse,…" - P22.L30-34 "Secondly, we incorporated in the staging of breast cancer patients. However, most of the patients are in phase II-III; some patients' staging is unclear. The treatment of patients receiving also is not the same, so we can't be used for clinical population-specific recommendations, but we think VR has significant benefits for patients with mental symptoms." -> This entire paragraph is unclear to me.

VERSION 2 – AUTHOR RESPONSE

Dear Dr. Thunyarat Anothaisintawee:

Thank you for your new comments on our paper. Upon receipt of your letter, we immediately began the modification work. We made a point-to-point reply here to explain our ideas and thoughts.

Method

• That is a good question! As you mentioned in your comments, our research work is too far from now. So we made a new database search, and our search results showed that there were no new literature available for inclusion.

• Thank you for pointing out the mistake. I think this is due to the difference in our language expression. What we want to say is that brain injury is associated with severe disease, not distant metastasis. We haven't ruled out distant metastases. Thank you for your valuable advice.

• We defined heterogeneity according to the Cochrane Library criteria and selected 50% as the risk of bias. Moreover, our research results show that there is almost no heterogeneity in one of our three forest maps, while the heterogeneity of the other two forest maps exceeds 50%. Therefore, we choose 50% as the high bias risk.

In short, we thank you for your new and valuable advice! Thank you also for your efforts in publishing this article.

Dear Dr. Katrin schoenenberg:

Thank you for your new round of review comments. We appreciate your comments and technical defects in this paper, and our gratitude is beyond words. It is thanks to peer review like yours that we can better promote academic exchange and progress.

• According to the standards of the Cochrane Library, random-effect models were selected for different types of trials. However, some studies were in the same category with low heterogeneity, so we chose fixed-effect models.

• Thanks for your question! This paper is a controlled trial before and after the intervention, comparing the difference before and after the intervention, and we do not need to calculate SMD, MD is the appropriate value. We highlighted it in this paper.

• Although there was no merger, the results of the before-after control trial showed that VR technology improved cognitive function, and there was a significant difference in the score after using VR, which we still believe is good for them.

We have revised all the grammar problems you mentioned in the document and have shown them in the article. Thanks for your efforts, we will do better and better.